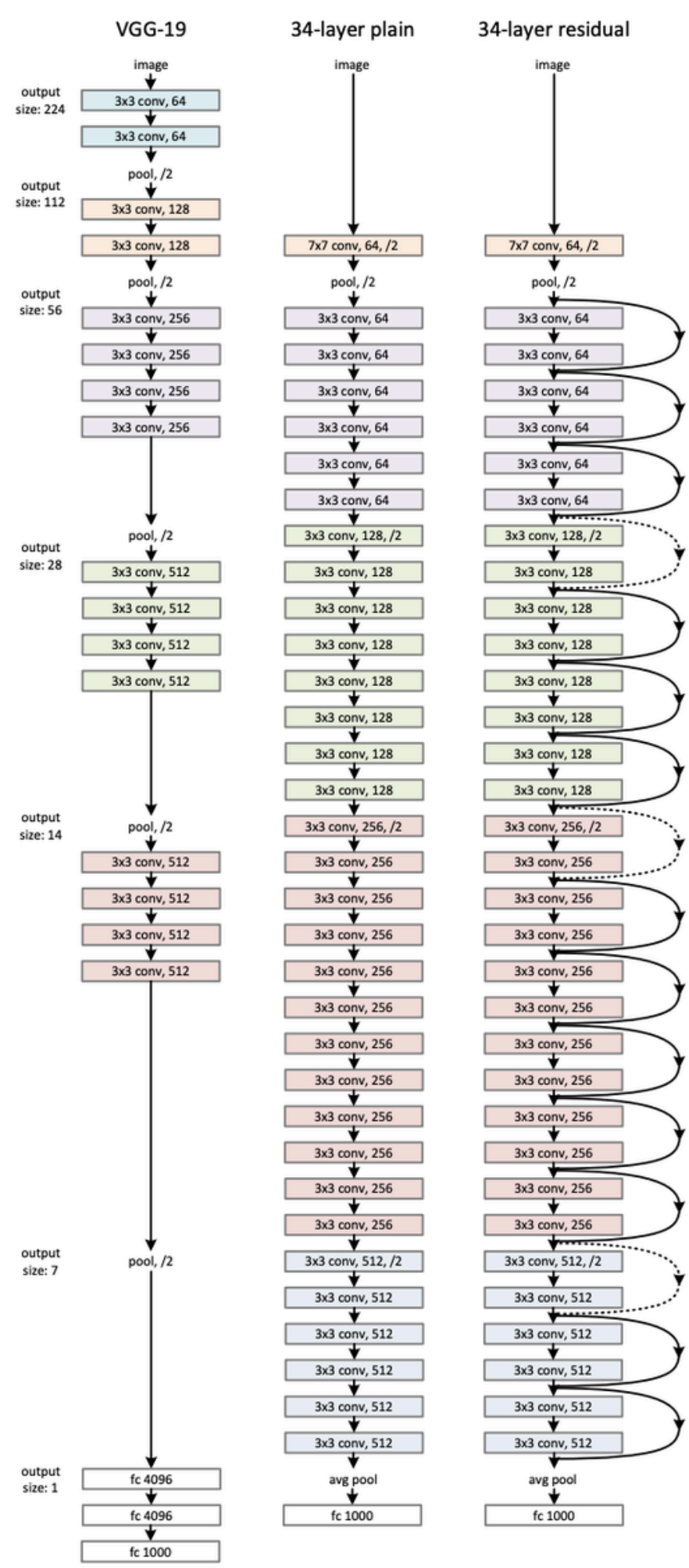
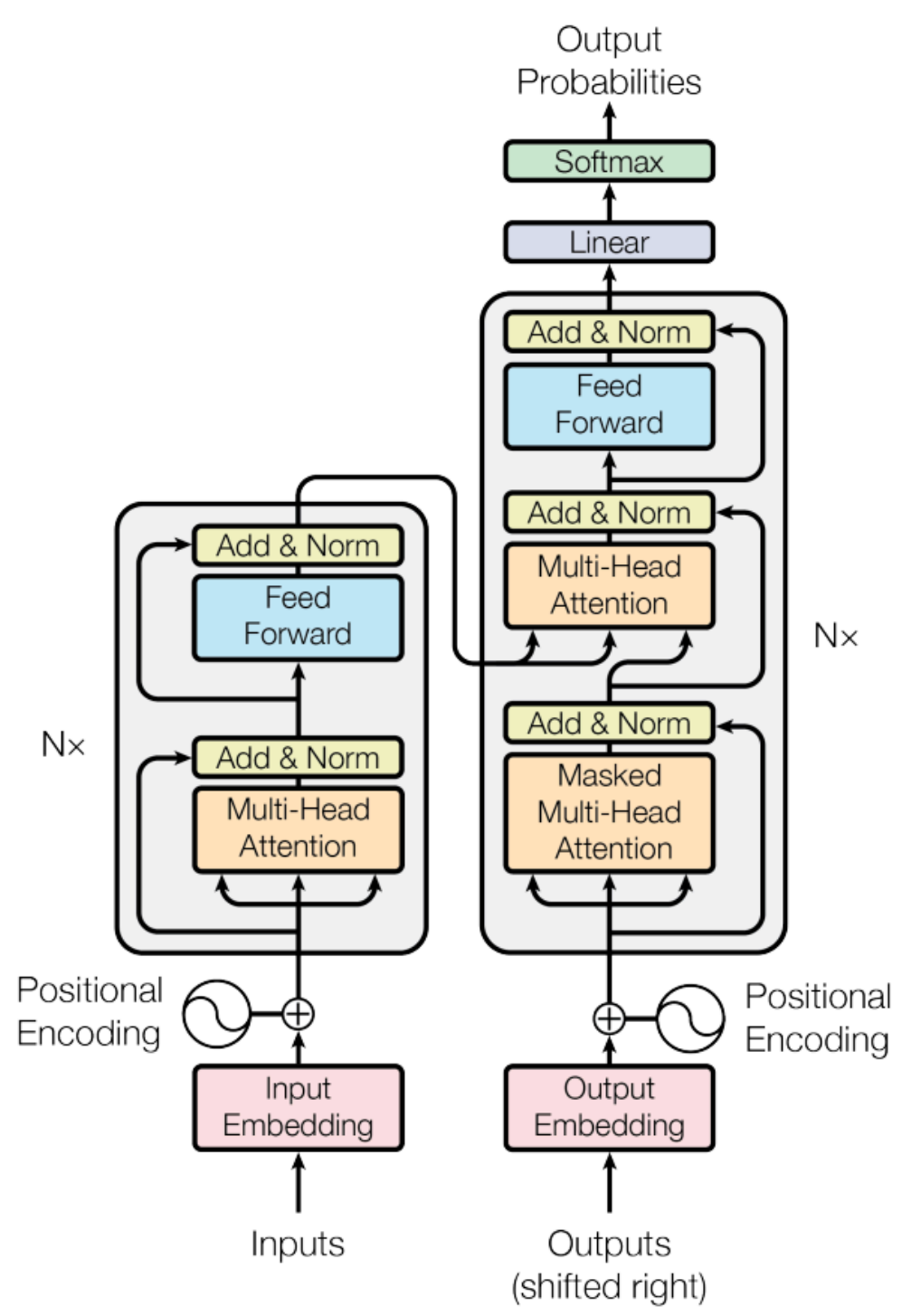


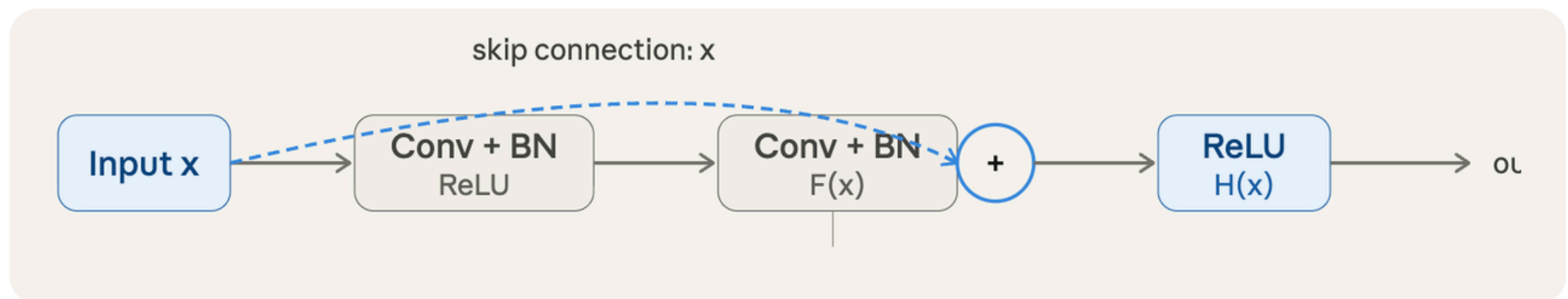
Best 5 Deep Learning Research Papers of All Time



Input	[CLS]	my	dog	is	cute	[SEP]	he	likes	play	##ing	[SEP]
Token Embeddings	$E_{[CLS]}$	E_{my}	E_{dog}	E_{is}	E_{cute}	$E_{[SEP]}$	E_{he}	E_{likes}	E_{play}	$E_{##ing}$	$E_{[SEP]}$
Segment Embeddings	E_A	E_A	E_A	E_A	E_A	E_A	E_B	E_B	E_B	E_B	E_B
Position Embeddings	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8	E_9	E_{10}



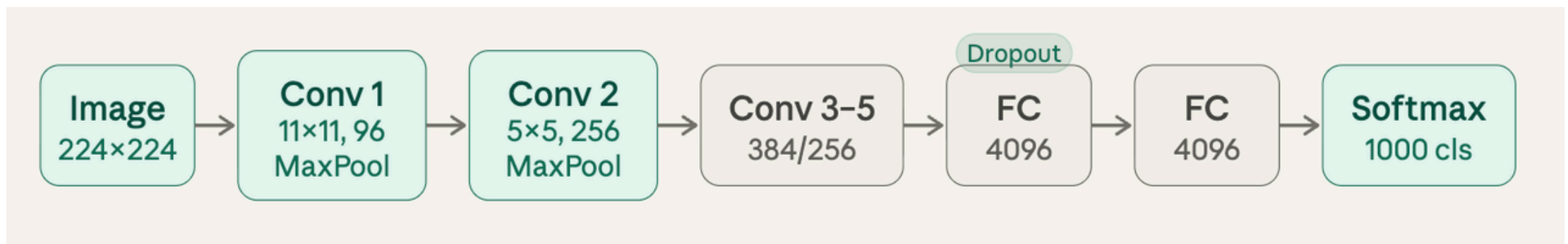
Deep Residual Learning for Image Recognition



Introduces residual (skip) connections that allow gradients to flow directly through shortcut paths, solving the vanishing gradient problem that had capped deep networks at ~20 layers. ResNet-152 achieved top-5 error of 3.57% on ImageNet surpassing human performance.

The core idea: instead of learning $H(x)$, layers learn the residual $F(x) = H(x) - x$.

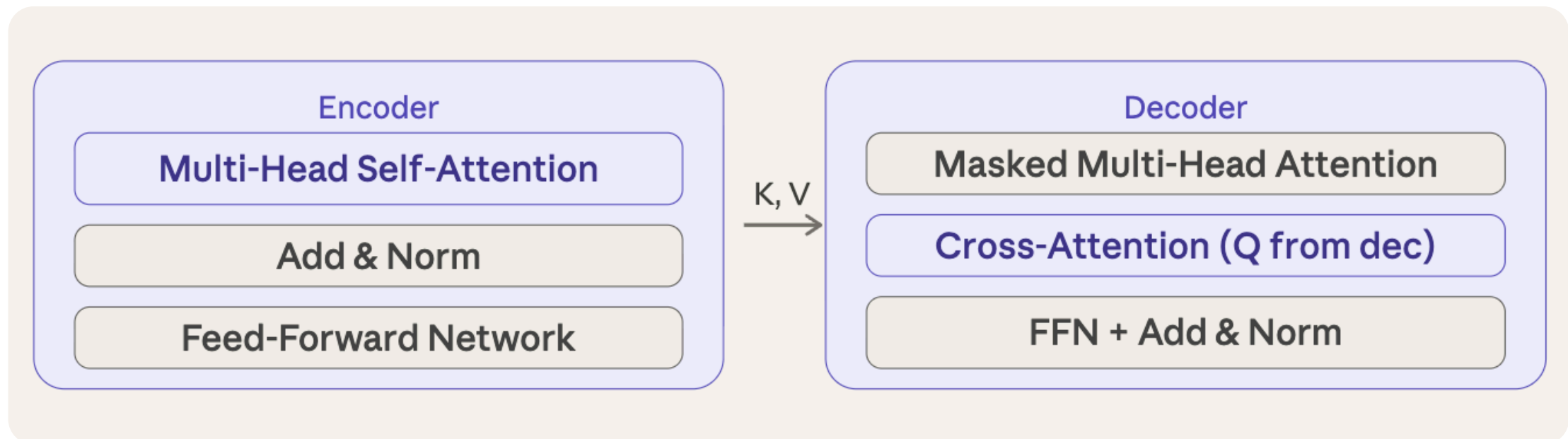
ImageNet Classification with Deep Convolutional Neural Networks



The paper that kicked off the deep learning revolution.

AlexNet trained an 8-layer CNN on GPUs and slashed ImageNet top-5 error from 26% to 15.3% a 10-point margin over the runner-up. It popularised ReLU activations, dropout regularisation, and data augmentation, setting the template for all subsequent CNN architectures.

Attention Is All You Need

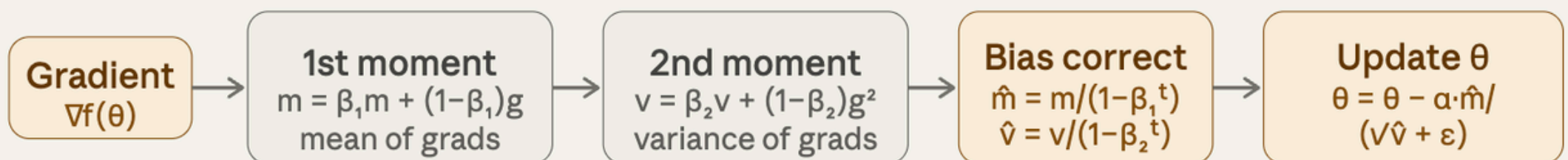


Introduces the Transformer architecture replacing recurrence entirely with multi-head self-attention.

Each token directly attends to every other token in the sequence, enabling full parallelism during training and capturing long-range dependencies that RNNs struggled with.

The Transformer is now the backbone of every major language model, from BERT to GPT to Claude.

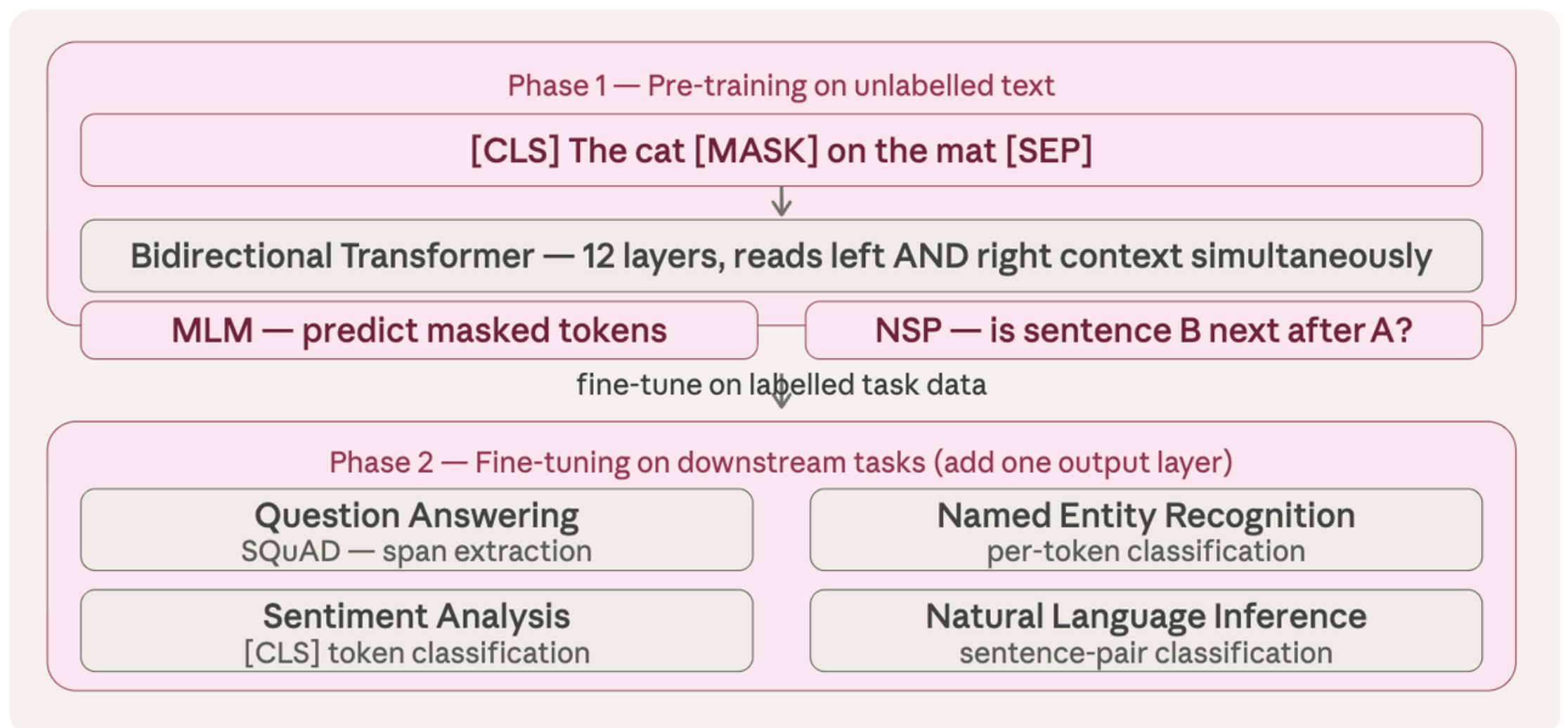
Adam: A Method for Stochastic Optimization



Introduces Adam (Adaptive Moment Estimation), combining the benefits of AdaGrad (per-parameter learning rates) and RMSProp (momentum tracking).

It maintains two running averages the first moment (mean of gradients) and the second moment (uncentred variance) with bias correction for early training steps. Adam is the default optimizer in almost every deep learning project today.

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding



Introduces BERT, a Transformer model pre-trained bidirectionally using Masked Language Modelling (MLM) and Next Sentence Prediction (NSP). Unlike earlier unidirectional LMs, BERT reads context from both left and right simultaneously. A single pre-trained BERT model can be fine-tuned with one extra layer to achieve state-of-the-art results on 11 NLP tasks including question answering, NLI, and named entity recognition.